

Task 1: To Create Main Loop, Display Prompt, Read User Input, Handle Exit Conditions, Design Control Flow Diagram, Test Interactive Loop

Task 2: Using the following system calls, copy the content from File1.txt to File2.txt.

1. **Open()**
2. **Read()**
3. **Write()**
4. **Close()**

Task 3: To Capture Keyboard Input, Handle Backspace, Process Enter Key, Manage Input Buffer, Support Multi-Character Commands, Test User Interaction.

Task 4: To Apply Escape Sequences, Store Command History, Navigate Previous Commands, Navigate Next Commands, Update Input Buffer, Test Recall Functionality

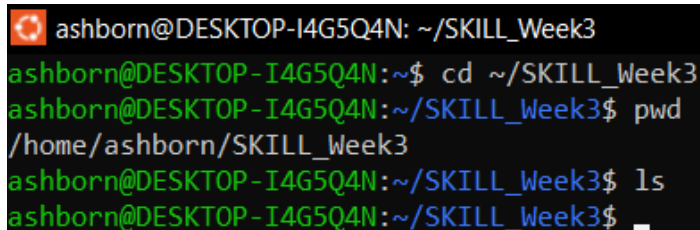
Task 5: To Allocate Buffers Dynamically, Resize Arrays, Prevent Buffer Overflow, Manage Linked Lists, Release Memory Correctly, Verify with Valgrind.

Validate the program with valgrind tool

MY SCREENSHOTS:

Task 1 –

1. Created a new folder for week 3



```
ashborn@DESKTOP-I4G5Q4N: ~/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~$ cd ~/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ pwd
/home/ashborn/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

2. Created task1.c and compiled successfully

```
ashborn@DESKTOP-I4G5Q4N:~$ cd ~/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ pwd
/home/ashborn/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task1.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task1.c -o task1
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

3. Executed task1 program

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task1.c -o task1
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task1
Enter a command (type exit to quit): hello
You entered: hello
Enter a command (type exit to quit): ubuntu
You entered: ubuntu
Enter a command (type exit to quit): ls
You entered: ls
Enter a command (type exit to quit): exit
Exiting program...
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

CONTROL FLOW DIAGRAM –

Start

|

Display Prompt

|

Read Input

|

Is Input = exit?

|

Yes → End

|

No

|

Print Input

|

Repeat Loop

TASK 2 –

1. Created a file and shown its content

```
Exiting program...
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano File1.txt
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ cat File1.txt
Hello Ubuntu
This is Week 3 Skill Lab
Learning System Calls
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

2. Created another file and compiled successfully

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task2.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task2.c -o task2
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

3. Copied file

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task2.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task2.c -o task2
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task2
File copied successfully.
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

4. Verified

```

ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano File1.txt
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ cat File1.txt
Hello Ubuntu
This is Week 3 Skill Lab
Learning System Calls
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task2.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task2.c -o task2
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task2
File copied successfully.
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ cat File2.txt
Hello Ubuntu
This is Week 3 Skill Lab
Learning System Calls

```

5. Final directory structure

```

ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls -l
total 33
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:42 File1.txt
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:51 File2.txt
-rwxr-xr-x 1 ashborn ashborn 16184 Aug  5 06:39 task1
-rw-r--r-- 1 ashborn ashborn  390 Aug  5 06:38 task1.c
-rwxr-xr-x 1 ashborn ashborn 16176 Aug  5 06:50 task2
-rw-r--r-- 1 ashborn ashborn  525 Aug  5 06:50 task2.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ █

```

TASK 3 –

1. We are in our required repository

```

ashborn@DESKTOP-I4G5Q4N:~$ cd ~/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ pwd
/home/ashborn/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
File1.txt File2.txt task1 task1.c task2 task2.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ █

```

2. Created and compiled task3 successfully

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task3.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task3.c -o task3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
File1.txt  File2.txt  task1  task1.c  task2  task2.c  task3  task3.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

3. Executed task3

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task3

mysh> hello
Command: hello

mysh> ls
Command: ls

mysh> nano
Command: nano

mysh> exit
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

4. List

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls -l
total 53
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:42 File1.txt
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:51 File2.txt
-rwxr-xr-x 1 ashborn ashborn 16184 Aug  5 06:39 task1
-rw-r--r-- 1 ashborn ashborn  390 Aug  5 06:38 task1.c
-rwxr-xr-x 1 ashborn ashborn 16176 Aug  5 06:50 task2
-rw-r--r-- 1 ashborn ashborn  525 Aug  5 06:50 task2.c
-rwxr-xr-x 1 ashborn ashborn 16232 Aug  7 08:17 task3
-rw-r--r-- 1 ashborn ashborn  1259 Aug  7 08:17 task3.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

5. Combo 1

```
ashborn@DESKTOP-I4G5Q4N:~$ cd ~/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ pwd
/home/ashborn/SKILL_Week3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
File1.txt File2.txt task1 task1.c task2 task2.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task3.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task3.c -o task3
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
File1.txt File2.txt task1 task1.c task2 task2.c task3 task3.c
```

6. Combo 2

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task3

mysh> hello
Command: hello

mysh> ls
Command: ls

mysh> nano
Command: nano

mysh> exit
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls -l
total 53
-rw-r--r-- 1 ashborn ashborn 60 Aug 5 06:42 File1.txt
-rw-r--r-- 1 ashborn ashborn 60 Aug 5 06:51 File2.txt
-rwxr-xr-x 1 ashborn ashborn 16184 Aug 5 06:39 task1
-rw-r--r-- 1 ashborn ashborn 390 Aug 5 06:38 task1.c
-rwxr-xr-x 1 ashborn ashborn 16176 Aug 5 06:50 task2
-rw-r--r-- 1 ashborn ashborn 525 Aug 5 06:50 task2.c
-rwxr-xr-x 1 ashborn ashborn 16232 Aug 7 08:17 task3
-rw-r--r-- 1 ashborn ashborn 1259 Aug 7 08:17 task3.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

TASK 4 –

1. Created and compiled the file

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task4.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task4.c -o task4
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
File1.txt  File2.txt  task1  task1.c  task2  task2.c  task3  task3.c  task4  task4.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

2. Executed the program successfully

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task4
Simple Command History
Type 'exit' to quit.

Command> hello
Stored: hello
History:
1. hello

Command> ubuntu
Stored: ubuntu
History:
1. hello
2. ubuntu

Command> linux
Stored: linux
History:
1. hello
2. ubuntu
3. linux

Command> exit
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

3. Final directory

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls -l
total 73
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:42 File1.txt
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:51 File2.txt
-rwxr-xr-x 1 ashborn ashborn 16184 Aug  5 06:39 task1
-rw-r--r-- 1 ashborn ashborn  390 Aug  5 06:38 task1.c
-rwxr-xr-x 1 ashborn ashborn 16176 Aug  5 06:50 task2
-rw-r--r-- 1 ashborn ashborn  525 Aug  5 06:50 task2.c
-rwxr-xr-x 1 ashborn ashborn 16232 Aug  7 08:17 task3
-rw-r--r-- 1 ashborn ashborn  1259 Aug  7 08:17 task3.c
-rwxr-xr-x 1 ashborn ashborn 16272 Aug  7 08:35 task4
-rw-r--r-- 1 ashborn ashborn  861 Aug  7 08:35 task4.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```

TASK 5 –

1. Created and compiled the file task5 successfully

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ nano task5.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc -task5.c -o task5
gcc: error: unrecognized command-line option '-task5.c'
gcc: fatal error: no input files
compilation terminated.
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ gcc task5.c -o task5
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls
File1.txt File2.txt task1 task1.c task2 task2.c task3 task3.c task4 task4.c task5 task5.c
```

2. Executed the program

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ./task5
Enter text: Hello Ubuntu
You entered: Hello Ubuntu
Hello Ubuntu - Memory Resized
Memory released successfully.
```


3. Installed valgrind successfully

```
ashborn@DESKTOP-I4G5Q4N:~$ sudo apt install valgrind
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libdw
Suggested packages:
  gdb-doc gdbserver valgrind-dbg valgrind-mpi kcachegrind alleyoop valkyrie
The following NEW packages will be installed:
  gdb libbabeltrace1 libc6-dbg libdebuginfod-common libdebuginfod1t64 libip
The following packages will be upgraded:
  libdw1t64 libelf1t64
2 upgraded, 9 newly installed, 0 to remove and 222 not upgraded.
Need to get 27.3 MB of archives.
After this operation, 103 MB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 libdebuginf
```

4. Executed the program using valgrind

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ valgrind ./task5
==454== Memcheck, a memory error detector
==454== Copyright (C) 2002-2022, and GNU GPL'd, by Julian Seward et al.
==454== Using Valgrind-3.22.0 and LibVEX; rerun with -h for copyright info
==454== Command: ./task5
==454==
==454== error calling PR_SET_PTRACER, vgdb might block
Enter text: Hello Ubuntu
You entered: Hello Ubuntu
Hello Ubuntu - Memory Resized
Memory released successfully.
==454==
==454== HEAP SUMMARY:
==454==      in use at exit: 0 bytes in 0 blocks
==454==    total heap usage: 4 allocs, 4 frees, 1,324 bytes allocated
==454==
==454== All heap blocks were freed -- no leaks are possible
==454==
==454== For lists of detected and suppressed errors, rerun with: -s
==454== ERROR SUMMARY: 0 errors from 0 contexts (suppressed: 0 from 0)
```

5. Final directory

```
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$ ls -l
total 93
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:42 File1.txt
-rw-r--r-- 1 ashborn ashborn  60 Aug  5 06:51 File2.txt
-rwxr-xr-x 1 ashborn ashborn 16184 Aug  5 06:39 task1
-rw-r--r-- 1 ashborn ashborn  390 Aug  5 06:38 task1.c
-rwxr-xr-x 1 ashborn ashborn 16176 Aug  5 06:50 task2
-rw-r--r-- 1 ashborn ashborn  525 Aug  5 06:50 task2.c
-rwxr-xr-x 1 ashborn ashborn 16232 Aug  7 08:17 task3
-rw-r--r-- 1 ashborn ashborn  1259 Aug  7 08:17 task3.c
-rwxr-xr-x 1 ashborn ashborn 16272 Aug  7 08:35 task4
-rw-r--r-- 1 ashborn ashborn  861 Aug  7 08:35 task4.c
-rwxr-xr-x 1 ashborn ashborn 16304 Aug  8 08:30 task5
-rw-r--r-- 1 ashborn ashborn  666 Aug  8 08:30 task5.c
ashborn@DESKTOP-I4G5Q4N:~/SKILL_Week3$
```